

Name: Pranjali Ingle Class: SOC-08 Batch: B
Subject: PL Roll No.: 43 Exp. No.: 1
Name of the Experiment: _____
Performed on: _____
Submitted on: _____

Marks	Teacher's Signature with date
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unit - 01 :- Introduction to Python Programming

Q1] List the key features and advantages of Python. Also mention any two Python IDEs or editors used for Program development.

→ Key features of Python are:-
1. Simple and easy to learn
2. Interpreted language
3. High level language
4. Object oriented
5. Portable.

★ Advantages of Python
1. Rapid development
2. Open source and there
3. Large community support.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

0 Two Python IDEs / Editors.

1 Pycharm 2. IDLE

Q.2] Explain the history and evolution of Python. Describe why Python has become popular. Industry applications.

→ Python was developed by Guido van Rossum in the 1980s and released in 1991. It was designed as a high-level interpreted language emphasizing code readability and simplicity.

0 Reasons for popularity of Python for Beginners

i] Simple and readable syntax

ii] Easy to learn and understand

iii] Interpreted language with easy debugging

0 Reasons for popularity of Python in Industry Applications

i] Versatile language used in multiple domains

ii] Rich set of libraries and framework

iii] Platform independent

- Q3] Write a Python Program to accept two numeric values from the user and perform the following
- 1] Addition
 - 2] Subtraction
 - 3] Multiplication
 - 4] Division.

Display the results using proper input and output prompting.

```
a = float(input("enter the first number:"))  
b = float(input("enter the second number:"))
```

```
addition = a+b
```

```
subtraction = a-b
```

```
multiplication = a*b
```

```
division = a/b
```

Displaying results

```
Print f ("Addition =", addition)
```

```
Print f ("Subtraction =", subtraction)
```

```
Print f ("Multiplication =", multiplication)
```

```
Print f ("Division =", division)
```

Q9]

Analyze the following Python code and identify the data types of each variable. Also explain the output.

→

```
a = 10
b = 5.5
c = 3+4j
d = True
```

Print (a+b, type(c), d)

Data types of Variables

a → int (integer)

b → float (floating point number)

c → complex (complex number with real and imaginary parts)

d → bool (Boolean value)

Output

a+b → 10 + 5.5 = 15.5

type(c) → Displays the data type of variable

which is < class 'complex' >

d → prints the Boolean value True.

15.5 < class 'complex' > True.

Q5] Evaluate the use of relational, logical, and bitwise operators in python. Justify with suitable examples where each type of operator is most appropriate.

Relational operator: Used for Comparison used in Conditional and decision Making

a = 10

b = 5

Print (a > b)

o Logical operators.

Used to combine Conditions.

Used in authentication and Validation

x = True

y = False

Print (x & y)

o Bitwise operators

operate on binary numbers used in low prog and networking

a = 5

b = 3

Print (a & b)

6] Design and implement a menu driven python Program that.

- o Accepts user input
- o Performs arithmetic, relational, logical and assignment operations
- o Display result clearly

The program should handle int float and boolean data type.

```
a = int (input ("enter first number:"))  
b = int (input ("enter second number:"))
```

```
Print ("1 Arithmetic")  
Print ("2 Relational")  
Print ("3 Logical")  
Print ("4 Assignment")
```

```
ch = int (input ("Enter choice:"))
```

```
if ch == 1;
```

```
    print ("Add = ", a+b)
```

```
    if
```

```
        ch = 2
```

```
        print ("a > b = ", a > b)
```

```
    else if
```

```
        ch = 4
```

```
        a + = b
```

```
        print ("a = ", a)
```

```
    else  
        print ("invalid choice")
```

7] Write a program in Python to display "Hello World"

Start

Print "Hello World"

End

Code

```
Print ("Hello World")
```

Output

Hello World

Algorithm

1. Start
2. Print Hello World
3. Stop.

e. Write a program to perform average of 5 numbers using input() function and type conversion concept.

```
a = float(input("enter number 1: "))  
b = float(input("enter number 2: "))  
c = float(input("enter number 3: "))  
d = float(input("enter number 4: "))  
e = float(input("enter number 5: "))
```

```
avg = (a + b + c + d + e) / 5  
print("average =", avg)
```

Output

```
enter number 1 = 10  
enter number 2 = 20  
enter number 3 = 30  
enter number 4 = 40  
enter number 5 = 50
```

Average = 30.0

Algorithm

1. Start
2. Accept 5 numbers from user
3. Calculate average
4. Display average
5. Stop

Start

Input a, b, c, d, e

$$\text{Avg} = (a + b + c + d + e) / 5$$

Print Avg

end

Q 9 Write a program that prompts user to enter their age and print out their age in years, months and days.

```
age = int(input("Enter age in years:"))  
Months = age * 12  
days = age * 365  
Print ("Age in months: ", months)  
Print ("Age in days: ", days)
```

Output

Enter Age in years : 18
Age in Months : 216
Age in days = 6570

Start

Input Age

Month = age * 12
days = age * 365

Algorithm

- 1 Start
 - 2 Input age in years
 - 3 calculate Months = age * 12
 - 4 calculate days = age * 365
 - 5 Display Month and days
 - 6 Stop
- Print Month, days
End.

Q10 Write a Program to demonstrate the concept of data types & conversion.

```
a=10  
b=float(a)  
c=str(a)
```

```
print(a, type(a))  
print(b, type(b))  
print(c, type(c))
```

Output

```
10 < class 'int' >  
10.0 < class 'float' >  
10 < class 'str' >
```

Algorithm.

- 1 Start
- 2 Assign integer Value
- 3 Convert integer to float and string
- 4 Display values and their data types
- 5 Stop

